Sir Syed CASE Institute of Technology

Visual Programming

Lecture 1: .Net Framework

3.00 (2 + 1) Credit Hours, Fall 2026

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Course outline

Visual studio

OOP in C#

Advance programming concepts in C#

Window form applications

Database Connections

ADO.NET

ASP.NET

Textbook

Microsoft Visual C# Step by Step

John Sharp

Course Details

Assignments → 7.5 %

Quizzes → 7.5%

Project → 10%

Mid Term → 20 %

Final Term → 35 %

Lab → 20 %

Introduction to Visual Studio

Integrated Development Environment

Developed by Microsoft

GUI (graphical user interface)

Web applications

Console applications

Independent of language specification

Python, php, C#, VB

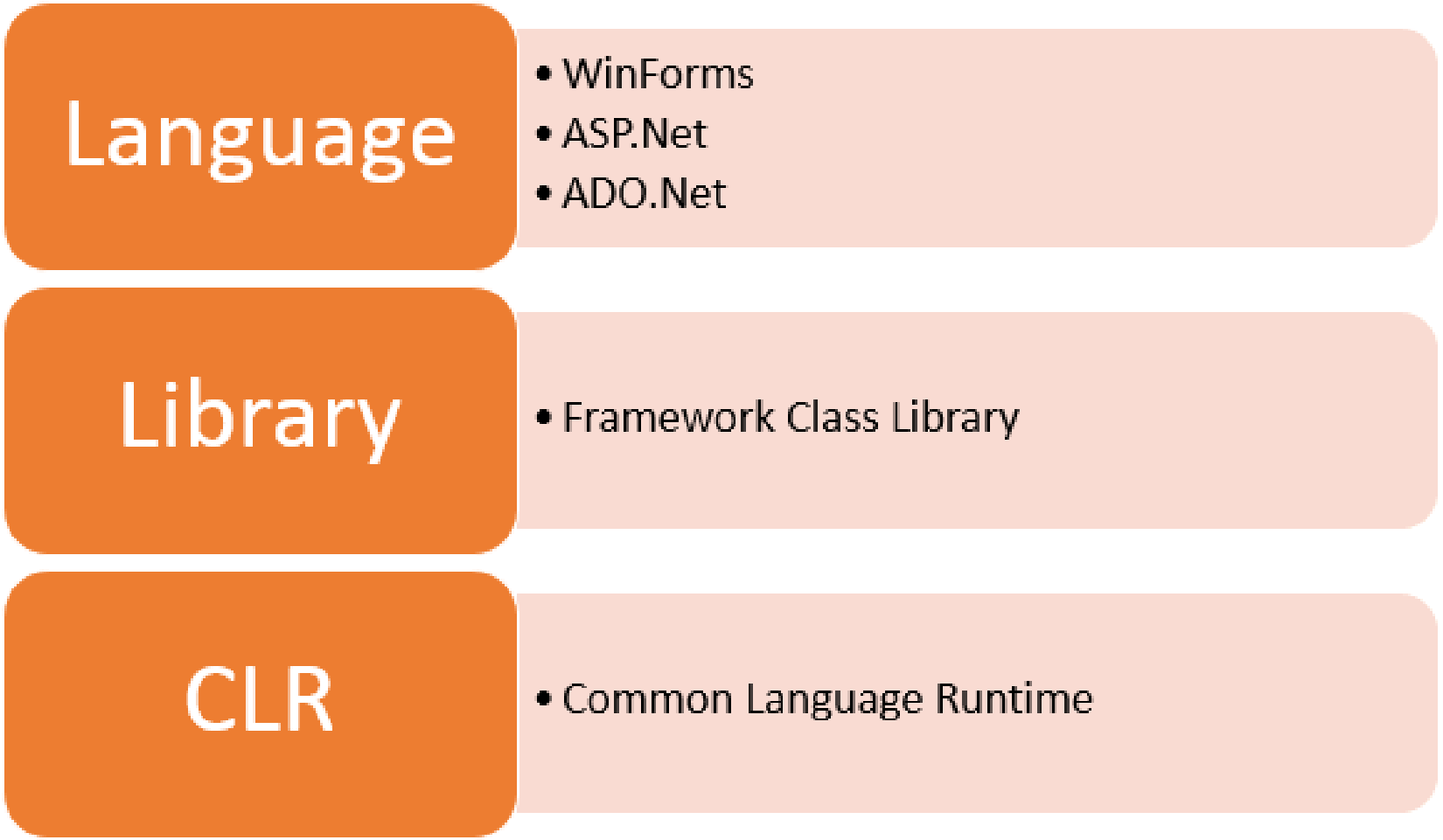
What is .Net Framework?

.Net Framework is a software development platform developed by Microsoft for building and running Windows applications.

The .Net framework consists of developer tools, programming languages, and libraries to build desktop and web applications.

It is also used to build websites, web services, and games.

.Net framework Architecture



.Net Framework Architecture

.Net Framework Architecture is a programming model for the .Net platform

Provides an execution environment and integration with various programming languages for simple development.

Deployment of various Windows and desktop applications.

It consists of class libraries and reusable components.

.Net Components

Common Language Runtime

- Helps to run code and provide services
- Design components and apps across different languages
- Objects written in different language can communicate
- For example: a class is written in different language and derived is written in another (libraries)
- Common type system defined by runtime
- Follow runtime rules for defining new types

.Net Components

Garbage Collection - Garbage collection is the process of removing unwanted resources when they are no longer required. Examples of garbage collection are

A File handle which is no longer required. If the application has finished all operations on a file, then the file handle may no longer be required.

The database connection is no longer required. If the application has finished all operations on a database, then the database connection may no longer be required.

.Net Components

Garbage Collection

- Automatic memory management
- Example : Object creation and release of memory after work is done
- Optimization engine
- Explicit call to garbage collector
  - Low physical memory
  - Memory allocated to various objects and threshold increased in heap
  - GC. Collect method used to perform garbage collection in unusual situation
  - Otherwise auto garbage collector is enough.

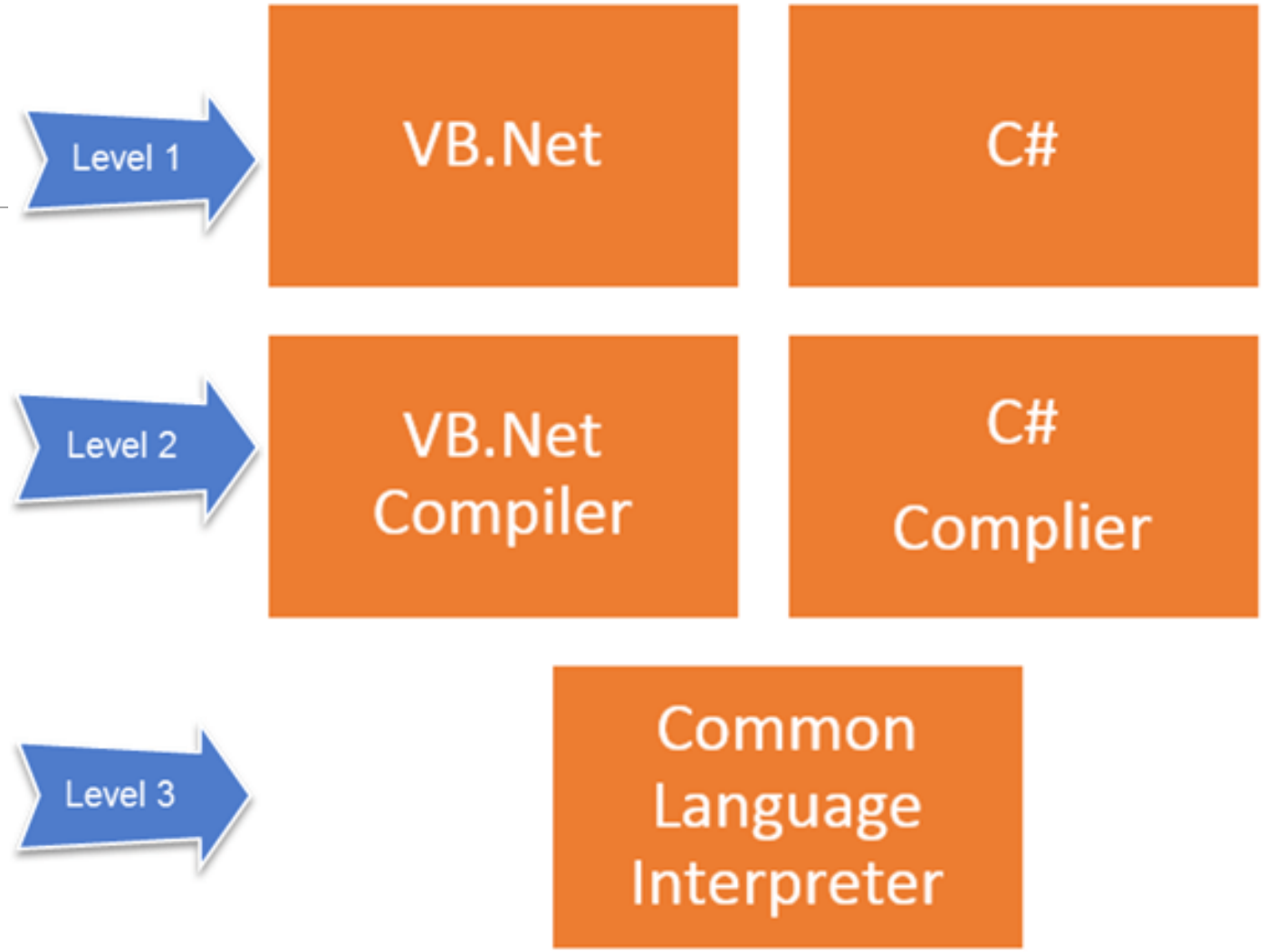
.Net Components

As noted in an earlier section, a developer can develop an application in a variety of .Net programming languages.

Language - The first level is the programming language itself, the most common ones are VB.Net and C#.

Compiler – There is a compiler which will be separate for each programming language. So underlying the VB.Net language, there will be a separate VB.Net compiler. Similarly, for C#, you will have another compiler.

Common Language Interpreter – This is the final layer in .Net which would be used to run a .net program developed in any programming language. So the subsequent compiler will send the program to the CLR layer to run the .Net application.


Level 1
VB.Net
C#
Level 2
VB.Net Compiler
C# Complier
Level 3
Common Language Interpreter

.Net Components

Class Library

The .NET Framework includes a set of standard class libraries.

A class library is a collection of methods and functions that can be used for the core purpose.

Collection of classes, namespaces, interfaces etc.

.Net Components

Supports following functions:

- Base and user defined data types
- Exception handling
- Input/output stream operation
- Communication
- GUI
- Web client and server
- Web services

.Net Components

Languages

The types of applications that can be built in the .Net framework is classified broadly into the following categories.

WinForms – This is used for developing Forms-based applications, which would run on an end user machine. Notepad is an example of a client-based application.

.NET Components

ASP.Net – This is used for developing web-based applications, which are made to run on any browser such as Internet Explorer, Chrome or Firefox.

The Web application would be processed on a server, which would have Internet Information Services Installed.

Internet Information Services or IIS is a Microsoft component which is used to execute an Asp. Net application.

The result of the execution is then sent to the client machines, and the output is shown in the browser.

ADO.Net – This technology is used to develop applications to interact with Databases such as Oracle or Microsoft SQL Server.

.Net Framework Design Principle

Interoperability - The .Net framework provides a lot of backward support. Suppose if you had an application built on an older version of the .Net framework, say 2.0. And if you tried to run the same application on a machine which had the higher version of the .Net framework, say 3.5. The application would still work. This is because with every release, Microsoft ensures that older framework versions get well with the latest version.

.Net Framework Design Principle

Portability- Applications built on the .Net framework can be made to work on any Windows platform. And now in recent times, Microsoft is also envisioning to make Microsoft products work on other platforms, such as iOS and Linux.

.Net Framework Design Principle

Security - The .NET Framework has a good security mechanism. The inbuilt security mechanism helps in both validation and verification of applications. Every application can explicitly define their security mechanism. Each security mechanism is used to grant the user access to the code or to the running program.

.Net Framework Design Principle

Memory management - The Common Language runtime does all the work or memory management. The .Net framework has all the capability to see those resources, which are not used by a running program. It would then release those resources accordingly. This is done via a program called the "Garbage Collector" which runs as part of the .Net framework. The garbage collector runs at regular intervals and keeps on checking which system resources are not utilized, and frees them accordingly.

.Net Framework Design Principle

Simplified deployment - The .Net framework also have tools, which can be used to package applications built on the .Net framework. These packages can then be distributed to client machines. The packages would then automatically install the application.

Managed Vs Unmanaged Code

.NET supports two kind of coding

Managed Code

Unmanaged Code

Managed Code

The resource, which is with in your application domain is, managed code. The resources that are within domain are faster.

The code, which is developed in .NET framework, is known as managed code. This code is directly executed by CLR with help of managed code execution. Any language that is written in .NET Framework is managed code.

Managed code uses CLR which in turns looks after your applications by managing memory, handling security, allowing cross - language debugging, and so on.

Unmanaged Code

The code, which is developed outside .NET, Framework is known as unmanaged code.

Applications that do not run under the control of the CLR are said to be unmanaged, and certain languages such as C++ can be used to write such applications, which, for example, access low - level functions of the operating system. Background compatibility with code of VB, ASP are examples of unmanaged code.

Unmanaged code can be unmanaged source code and unmanaged compile code.

Summary

.Net Framework

.Net Framework architecture and components

Design principles of .Net

Managed vs unmanaged code